



RU

2021

1 (a)

▼ Define the order and degree of a differential equation with examples #2016 #2017 #2019 #2021.

Find the differential equation of all circles passing through the origin and having their centers on the x-axis

The degree of a differential equation is the highest power of the highest order derivative present in the differential equation. For example, $[y'' + y']^4 = k^2(y''')^2$

The highest derivative present in a differential equation is called the order of a differential equation. For example, $y'' + 3xy' + 4y = \sin(x)$. The order of this differential equation is two.

Let, the equation of the circle assuming R as radius and (a, b) as the centre of the circle is,

$$(x - a)^2 + (y - b)^2 = R$$

Since, the radius of the circle passes through the origin with the centre on x-axis, so the centre of circle will be (a, 0).

Thus, equation of the circle is,

$$(x - a)^2 + y^2 = a^2 \quad \rightarrow (1)$$

Differentiating both sides we get,

$$2(x - a)dx + 2ydy = 0$$

$$\Rightarrow x - a = -\frac{ydy}{dx}$$

$$\text{and } a = x + \frac{ydy}{dx}$$

Substituting these values in equation (1),

$$\left(-\frac{ydy}{dx}\right)^2 + y^2 = \left(x + \frac{ydy}{dx}\right)^2$$

$$\Rightarrow y^2 \left(\frac{dy}{dx}\right)^2 + y^2 = x^2 + y^2 \left(\frac{dy}{dx}\right)^2 + 2xy \frac{dy}{dx}$$

$$\Rightarrow y^2 = x^2 + y^2 \left(\frac{dy}{dx}\right)^2$$

Hence, this is the required differential equation of all the circles passing through the origin and having their centres on the x-axis.

1 (b)

▼ Form the differential equation of the family of parabolas with focus at the origin and axes along the x-axis

Standard equation of a parabola that opens up and symmetric about x-axis with the vertex (h, k) .

$$(y-k)^2 = 4a(x-h)$$

Now, Differentiate —

$$2(y-k) y' = 4a$$

$$y' = \frac{2a}{y-k} \Rightarrow y' = \frac{2a}{(y-k)}$$

$$(y-k) = \frac{2a}{y'}$$

Now, Again Differentiate —

$$y'' = \frac{2ay'}{(y-k)^2}$$

Putting $(y-k) = \frac{2a}{y'}$, then.

$$y'' = \frac{2ay'}{\left(\frac{2a}{y'}\right)^2}$$

$$y'' = \frac{2ay'}{\frac{4a^2}{(y')^2}}$$

$$y'' = \frac{(y')(y')^2}{2a}$$

$$2ay'' = (y')^3$$

$$(y')^3 - 2ay'' = 0$$

Hence, the differential equation of the family of parabolas with axis parallel to the x axis and with the distance from vertex to focus fixed as 'a' is-

$$(y')^3 - 2a'' = 0$$

1 (c)

▼ Solve $y = x^2 p^2 - px$ where $p \equiv \frac{dy}{dx}$ #2019 #2018 #2017

First, let's differentiate both sides of the equation $y = x^2 p^2 - px$ with respect to x :

$$\frac{d}{dx} y = \frac{d}{dx} (x^2 p^2) - \frac{d}{dx} (px)$$

Using the chain rule on the left-hand side and the product rule on the right-hand side, we get:

$$\frac{dy}{dx} = 2xp^2 + 2x^2 p \frac{dp}{dx} - p - x \frac{dp}{dx}$$

Now, we can substitute the given condition $p \equiv \frac{dy}{dx}$ into this expression:

$$p \equiv \frac{dy}{dx} = 2xp^2 + 2x^2 p \frac{dp}{dx} - p - x \frac{dp}{dx}$$

Simplifying this equation, we get:

$$(2xp - 1)p = x \frac{dp}{dx} (2xp - 1)$$

Dividing both sides by $(2xp - 1)x$ (assuming $2xp - 1 \neq 0$), we obtain:

$$\frac{p}{x} - \frac{1}{2xp - 1} = \frac{d}{dx} (\ln |x|)$$

This is a separable differential equation which can be integrated to find $p(x)$:

$$\int \left(\frac{p}{x} - \frac{1}{2xp - 1} \right) dx = \int \frac{d}{dx} (\ln |x|) dx$$

$$\ln |x| - \ln |2xp - 1| = \ln |c|$$

$$|x| = c|2xp - 1|$$

where c is an arbitrary constant. Solving for p , we get:

$$p = \frac{1}{2x} (1 \pm \sqrt{1 - 4cx})$$

Now we can substitute this expression for p back into the original equation $y = x^2 p^2 - px$ to obtain the solution for $y(x)$ in terms of c :

$$y(x) = \frac{1}{4c^2} ((1 - 4cx)^2 + 4cx) + \frac{\ln |x|}{2c} + \frac{1}{4c}$$

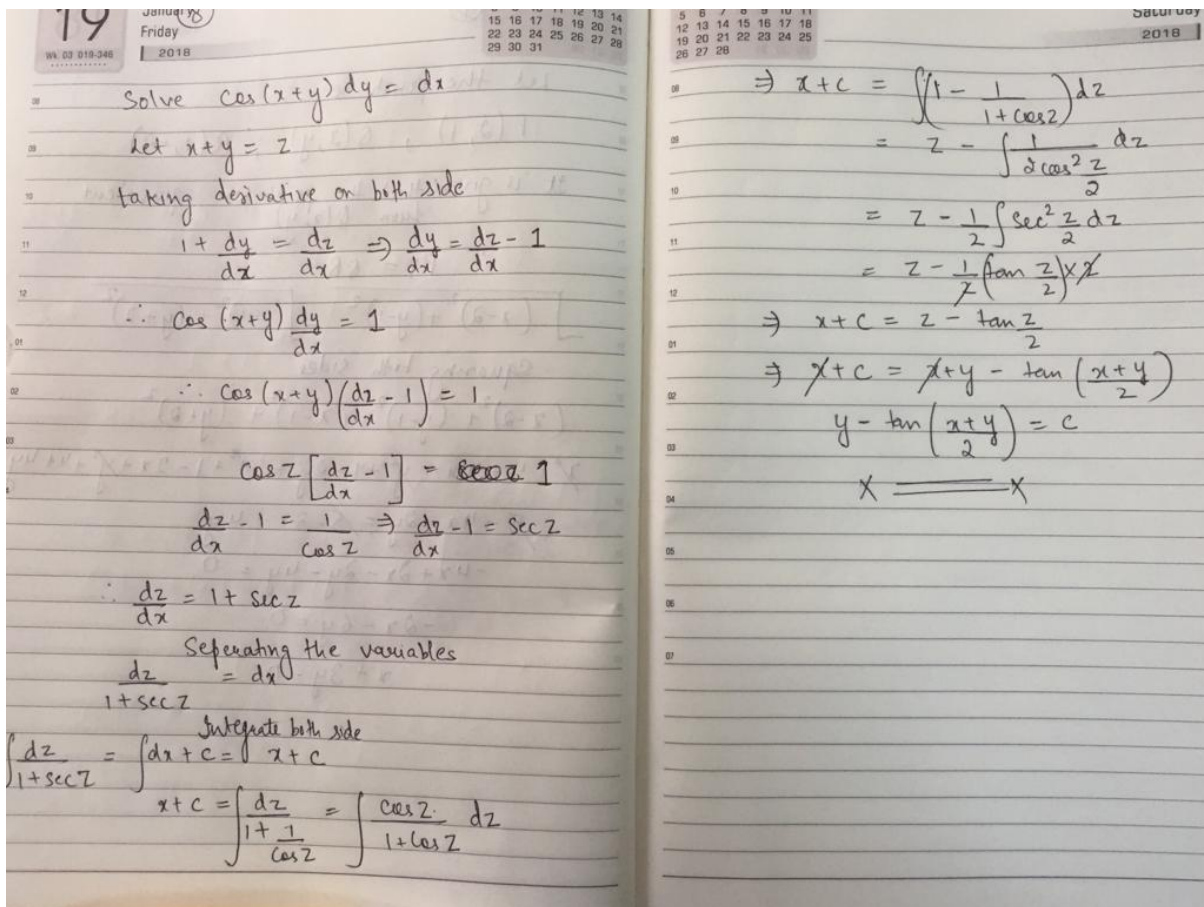
Therefore, the general solution to the differential equation $y = x^2 p^2 - px$ is

$$y(x) = \frac{1}{4c^2} ((1 - 4cx)^2 + 4cx) + \frac{\ln |x|}{2c} + \frac{1}{4c},$$

where c is an arbitrary constant.

2 (a)

▼ **Solve** $\cos(x + y)dy = dx$ **#review-1**



2 (b)

▼ Solve $\frac{dy}{dx} = x^3 y^3 - xy$ #2018

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/6dd6fb65-5549-4e77-a82d-2167a590b3e6/2021_2b.pdf

2 (c)

▼ Define Bernoulli differential equation #2021 #2020 #2019 #2017. Solve $\frac{dy}{dx} + y = xy^3$ #2021 #2020 #2019

A Bernoulli differential equation is a type of first-order ordinary differential equation that can be written in the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

What is the general solution of the differential equation $dy/dx + y = xy^3$? | Socratic

See below. Making the change of variable $y = 1/z$ we have the new version $dy/dx + y - xy^3 = 0$

$\rightarrow (x - z^2 + z z')/z^4 = 0$ or $x - z^2 + z z' = 0$ Now calling $xi = z^2$ we have $x - xi + 1/2 xi' = 0$

Solving for xi we obtain easily $xi = 1/2 + x + C e^{(2x)} = z^2$ then $z = \pm \sqrt{1/2 + x + C e^{(2x)}}$

<https://socratic.org/questions/5967a14fb72cff2ef05dabe8>



https://s3-us-west-2.amazonaws.com/secure.notion-static.com/0589c7ef-43c6-4fe7-98b1-0d0043802f8e/2021_2c.pdf

or accurate one

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/324d4eaa-f0fe-4e86-8619-958de4b9216b/2021_2c_short.pdf

3 (a)

▼ Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\cos x$ by using method of undetermined coefficients

Ex. 4. Using the method of undetermined coefficients to solve $(d^2y/dx^2) - 2(dy/dx) - 3y = 2e^x - 10\sin x$. [Delhi Maths Hons. 1997]

Sol. Given $(D^2 - 2D - 3)y = 2e^x - 10\sin x$, where $D \equiv d/dx$ (1)

Its auxiliary equation is $D^2 - 2D - 3 = 0$ so that $D = -1, 3$.

$\therefore$ C.F. $= c_1 e^{-x} + c_2 e^{3x}$, c_1, c_2 being arbitrary constants. ... (2)

Let the trial solution be $y^* = A e^x + B \sin x + C \cos x$ (3)

Since y^* must satisfy (1), $D^2 y^* - 2D y^* - 3 y^* = 2 e^x - 10 \sin x$ (4)

From (3), $D y^* = A e^x + B \cos x - C \sin x$, $D^2 y^* = A e^x - B \sin x - C \cos x$... (5)

Using (3) and (5), (4) reduces to

$$(A e^x - B \sin x - C \cos x) - 2(A e^x + B \cos x - C \sin x) - 3(A e^x + B \sin x + C \cos x) = 2e^x - 10 \sin x$$

$$\text{or } -4 A e^x - (4B - 2C) \sin x - (4C + 2B) \cos x = 2e^x - 10 \sin x.$$

Equating the coefficients of like terms in above identity, we have

$$-4A = 2, \quad -(4B - 2C) = -10 \quad \text{and} \quad -(4C + 2B) = 0 \quad \Rightarrow \quad A = -1/2, \quad B = 2, \quad C = -1$$

$$\therefore \text{ From (3), } y^* = (-1/2) e^x + 2 \sin x - \cos x.$$

$$\text{and general solution is } y = \text{C.F.} + y^* \text{ i.e., } y = c_1 e^{-x} + c_2 e^{3x} - (1/2) e^x + 2 \sin x - \cos x.$$

To solve this differential equation by using the method of undetermined coefficients, we assume that the particular solution has the form:

$$y_p = Ae^x + B \cos x + C \sin x + D$$

where A , B , C and D are constants to be determined.

Taking the first and second derivatives of y_p with respect to x gives:

$$\frac{dy_p}{dx} = Ae^x - B \sin x + C \cos x$$

$$\frac{d^2y_p}{dx^2} = Ae^x - B \cos x - C \sin x$$

Substituting these into the original differential equation, we get:

$$(Ae^x - B \cos x - C \sin x) - 2(Ae^x - B \sin x + C \cos x) - 3(Ae^x + B \cos x + C \sin x + D) = 2e^x - 10 \cos x$$

Simplifying this equation, we get:

$$(A - 2B - 3A)e^x + (C - 2B - 3B) \cos x + (-C + 2A - 3D) \sin x - 3D = 2e^x - 10 \cos x$$

Comparing the coefficients of like terms on both sides of the equation, we obtain the following system of equations:

$$\begin{aligned} A - 5e^x &= 0 \\ -2B - 10 \cos x &= -10 \cos x \\ -2C &= 0 \\ -3A - 3B - 3C - 3D &= -3D + 2e^x \end{aligned}$$

Solving for the constants, we get:

$$A = 5e^x$$

$$B = 0$$

$$C = 0$$

$$D = \frac{2}{3}e^x - \frac{2}{9}$$

Therefore, the general solution to the differential equation is:

$$y = y_h + y_p = c_1 e^{-x} + c_2 e^{3x} + 5e^x + \frac{2}{3}e^x - \frac{2}{9}$$

where y_h is the homogeneous solution.

3 (b)

▼ Solve the differential equation $(D^2 - 4D + 4)y = 3x^2 e^{2x} \sin 2x$ by operator method

We have

$$(D^2 - 4D + 4)y = 3x^2 e^{2x} \sin 2x$$

$$\text{A.E. is } m^2 - 4m + 4 = 0$$

$$\text{i.e., } (m - 2)^2 = 0$$

$$m = 2, 2$$

$$\text{C.F.} = (C_1 + C_2 x) e^{2x}$$

$$\text{P.I.} = \frac{1}{(D - 2)^2} \cdot 3x^2 e^{2x} \sin 2x$$

Taking e^{2x} outside the operator
and changing D to $D + 2$

$$= 3e^{2x} \frac{1}{D^2} x^2 \sin 2x$$

We shall find $\frac{1}{D^2} x^2 \sin 2x$

Integrating twice

$$\frac{1}{D} (x^2 \sin 2x) = \int x^2 \sin 2x dx$$

Applying Integration by parts

$$\begin{aligned} &= x^2 \left(\frac{-\cos 2x}{2} \right) - 2x \left(\frac{-\sin 2x}{4} \right) + 2 \left(\frac{\cos 2x}{8} \right) \\ &= \frac{-x^2 \cos 2x}{2} + \frac{x \sin 2x}{2} + \frac{\cos 2x}{4} \end{aligned}$$

Hence, $\frac{1}{D^2} (x^2 \sin 2x)$

$$= \frac{-1}{2} \int x^2 \cos 2x dx + \frac{1}{2} \int x \sin 2x dx + \frac{1}{4} \int \cos 2x dx$$

$$\begin{aligned} &= \frac{-1}{2} \left[x^2 \frac{\sin 2x}{2} - 2x \left(\frac{-\cos 2x}{4} \right) + 2 \left(\frac{-\sin 2x}{8} \right) \right] \\ &\quad + \frac{1}{2} \left[x \left(\frac{-\cos 2x}{2} \right) - 1 \left(\frac{-\sin 2x}{4} \right) \right] + \frac{1}{4} \cdot \frac{\sin 2x}{2} \end{aligned}$$

$$= \frac{-1}{8} (2x^2 \sin 2x + 4x \cos 2x - 3 \sin 2x)$$

$$\therefore \text{P.I.} = \frac{-3}{8} e^{2x} (2x^2 \sin 2x + 4x \cos 2x - 3 \sin 2x)$$

$\therefore$ The general solution is

$$y = \text{C.F.} + \text{P.I.}$$

$$\begin{aligned} &= (C_1 + C_2 x) e^{2x} - \frac{3}{8} e^{2x} (2x^2 \sin 2x \\ &\quad + 4x \cos 2x - 3 \sin 2x). \end{aligned}$$

3 (c)

▼ Solve $(D^2 + 4)y = \cos^2 x$ #2021 #2017

https://www.sarthaks.com/?qa=blob&qa_blobid=17652958957387634343

or

$$(D^2 + 4)y = \cos(2x)$$

Complementary Function

Auxiliary equation is

$$D^2 + 4 = 0$$

$$D^2 = -4$$

$$D = \pm 2i$$

$$y = A \cos(2x) + B \sin(2x)$$

Particular Integral

$$y = \frac{1}{D^2 + 4} \cos(2x)$$

$$\text{Put } y = \frac{1}{f(D)} \cos ax$$

Replace D^2 by $-a^2$

If it is zero then

$$y_2 = x \frac{1}{f'(x)} \cdot \cos(ax)$$

Now Replace D^2 in $f'(x)$ by $-a^2$

$$\therefore y_2 = \frac{1}{D^2 + 4} \cos(2x)$$

$$= x \cdot \frac{1}{2D} \cos(2x)$$

$$= \frac{x}{2} \int \cos(2x) \cdot dx$$

$$= \frac{x}{4} \sin(2x)$$

Complete solution is

$$y = A \cos 2x + B \sin 2x + \frac{x}{4} \sin(2x)$$

$$y = A \cos 2x + \left(B + \frac{x}{4}\right) \sin(2x)$$

4 (a)

▼ Define regular singular point #2021 #2020 #2017 #2016. Find the general solution $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 + 2)y = 0$ in power of x about $x_0 = 0$

a regular singular point of a linear differential equation is a point where the equation can be expressed in a specific form that allows for a solution in terms of a power series. Specifically, a point x_0 is said to be a regular singular point of a linear differential equation if the equation can be written in the form:

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = 0,$$

4 (b)

▼ **Write Helmholtz's equation and solve it #2016**

https://www.youtube.com/watch?v=TiF6D2rCOHQ&ab_channel=NeelmaniGupta

Helmholtz Equation:

Helmholtz equation is used in physics and mathematics. It is a partial differential equation and its mathematical formula is -

$$\nabla^2 A + k^2 A = 0$$

where, ∇^2 : Laplacian

k : wavenumber

A : amplitude

Helmholtz equation finds application in physics problem solving and electromagnetic radiation.

Helmholtz Equation Derivation:

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) U(x, t) = 0 \text{ (wave equation)}$$

$$U(x, t) = A(x) T(t) \text{ (separation of variables)}$$

$$\frac{\nabla^2 A}{A} = \frac{1}{c^2 T} \frac{d^2 T}{dt^2} \text{ (substitution into the wave equation)}$$

$$\frac{\nabla^2 A}{A} = -k^2$$

$$\text{and } \frac{1}{c^2 T} \frac{d^2 T}{dt^2} = -k^2 \text{ (above two are obtained equations)}$$

$$\nabla^2 A + k^2 A = 0 \text{ (Helmholtz equation after rearranging)}$$

This was the Helmholtz equation soln.

Application: There are three types of seismic waves: body waves which have p-waves (primary waves) and s-waves (secondary waves), surface waves and normal waves.

Seismology: The scientific study of earthquakes and their elastic waves in seismology.

5 (a)

- ▼ What is an optimization problem? Describe unconstrained and constrained optimization problems
1. **Optimization Problem:** An optimization problem involves finding the best solution (maximum or minimum) from a set of possible solutions, subject to certain constraints and objective functions.
 2. **Unconstrained Optimization:** In unconstrained optimization, the objective function is to be optimized without any restrictions on the variables. The goal is to find critical points where the derivative is zero to identify the maximum or minimum values.
 3. **Constrained Optimization:** Constrained optimization deals with optimizing a function subject to specific constraints. The objective is to find critical points considering both the objective function and the constraints. Methods like Lagrange multipliers are used to solve these problems.

5 (b)

- ▼ Define: linear programming problem, nonlinear programming problem, and convex optimization problem

Linear Programming Problem: Linear programming is an optimization technique used to find the best outcome in a mathematical model with linear relationships.

Nonlinear Programming Problem: Nonlinear programming deals with optimizing functions that involve nonlinear relationships between the variables.

Convex Optimization Problem: Convex optimization involves optimizing a convex objective function over a convex set of feasible solutions

5 (c)

- ▼ What do you mean by minimizer of a function? Define local and global minimizer of a function f . **Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = (x + 1)^2 + 3$ find the arg min $f(x)$**

Minimizer of a Function:

A minimizer of a function refers to a value that produces the smallest output of the function among a specific set of inputs.

Local Minimizer of a Function:

A local minimizer of a function f is a point 'a' in the domain of f such that $f(a)$ is less than or equal to $f(x)$ for all x in some open interval containing 'a'.

Global Minimizer of a Function:

A global minimizer of a function f is a point 'b' in the domain of f such that $f(b)$ is less than or equal to $f(x)$ for all x in the entire domain of f

6 (a)

- ▼ Define feasible direction of a vector and directional derivative of a function f . Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x) = x^T \begin{pmatrix} \frac{1}{4} & \frac{2}{7} \end{pmatrix} x + x^T \begin{pmatrix} \frac{3}{5} \end{pmatrix} + 6$. Find the directional derivative of f at $(1, 1)^T$ with respect

to a unit vector in the direction of maximal rate of increase

a feasible direction of a vector represents a direction in which the vector can be varied or scaled while still satisfying certain constraints.

The directional derivative of a function f in the direction of a vector v at a point x , denoted as $D_v f(x)$, represents the rate of change of the function f in the direction of v at the point x .

The function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined as:

$$f(x) = x^T \begin{bmatrix} \frac{1}{4} \\ \frac{2}{7} \end{bmatrix} x + x^T \begin{bmatrix} \frac{3}{5} \\ 6 \end{bmatrix} + 6.$$

$$v = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

$$\nabla f = \begin{bmatrix} \frac{1}{4} \\ \frac{2}{7} \end{bmatrix} x + \begin{bmatrix} \frac{3}{5} \\ 6 \end{bmatrix}.$$

$$D_v f(1, 1) = \nabla f^T(1, 1) \cdot v.$$

$$D_v f(1, 1) = \begin{bmatrix} \frac{1}{4} \\ \frac{2}{7} \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

The dot product is:

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

$$D_v f(1, 1) = \left(\frac{1}{4} + \frac{2}{7} \right) \cdot \left(\frac{1}{\sqrt{2}} \right) + \left(\frac{3}{5} + 6 \right) \cdot \left(\frac{1}{\sqrt{2}} \right).$$

$$= 5.045$$

6 (b)

▼ Define gradient and Hessian matrix of $f : \mathbb{R}^n \rightarrow \mathbb{R}$. State First-Order Necessary Condition. **Suppose there are two base station antennas, one for the primary base station and another for the neighboring base station. Both antennas are transmitting signals to the mobile user at equal power. However, the power of the received signal as measured by the mobile is the reciprocal of the squared distance from the associated antenna. Find the position of the mobile that maximizes the**

signal-to-interference ratio, which is the ratio of the received signal power from the primary base station to be received signal power from the neighboring base station

Gradient and Hessian Matrix:

The gradient of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a vector of partial derivatives, representing the rate and direction of steepest ascent of the function at a specific point. If f is differentiable, the gradient ∇f is given by:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}.$$

The Hessian matrix of f is a square matrix of second-order partial derivatives and provides information about the curvature of the function at a specific point. For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Hessian matrix H is defined as:

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

First-Order Necessary Condition:

For an optimization problem, the first-order necessary condition states that if a function f has a local maximum or minimum at a critical point x^* , and if f is differentiable at x^* , then the gradient of f at x^* must be the zero vector:

$$\nabla f(x^*) = 0.$$

6 (c)

▼ Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be the continuously differentiable function and $\{x_k\}$ be a sequence generated by Steepest descent algorithm. Prove that, for each k , the vectors $x_{k+1} - x_k$ is orthogonal to the vector $x_{k+2} - x_{k+1}$

6(c) If $[x_k]$ is a steepest descent sequence for $f: \mathbb{R}^n \rightarrow \mathbb{R}$ this for each k , the vector $x_{k+1} - x_k$ is orthogonal to the vector $x_{k+2} - x_{k+1}$

= Linear product symbol $(x_{k+1} - x_k, x_{k+2} - x_{k+1})$

~~$= \alpha_k \alpha_{k+1} \langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle$~~

$= \alpha_k \alpha_{k+1} \langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle$

$= 0$

7 (a)

▼ Define the epigraph of a function and a convex function. Let $f: \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function on Ω . Prove that Ω is a convex set

Epigraph of a function:

which is a subset of $\mathbb{R}^{n+1}$. The *epigraph* of a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is defined as

$$\text{epi } f = \{(x, t) \mid x \in \text{dom } f, f(x) \leq t\},$$

which is a subset of $\mathbb{R}^{n+1}$. ('Epi' means 'above' so epigraph means 'above the

Convex function:

A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is *convex* if $\text{dom } f$ is a convex set and if for all $x, y \in \text{dom } f$, and θ with $0 \leq \theta \leq 1$, we have

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y). \quad (3.1)$$

If a func $f: \Omega \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is convex on Ω , then show that Ω is convex. 7(a)

= Suppose, Ω is not a convex set.

Then for some $\alpha \in (0, 1)$ there exists two points y_1 and

y_2 such that, $z = \alpha y_1 + (1-\alpha)y_2 \notin \Omega$

Let, $\beta_1 = f(y_1)$, $\beta_2 = f(y_2)$

Let, the point $(y_1, \beta_1)^T, (y_2, \beta_2)^T$

$$\therefore w = \alpha \begin{pmatrix} y_1 \\ \beta_1 \end{pmatrix} + (1-\alpha) \begin{pmatrix} y_2 \\ \beta_2 \end{pmatrix}$$

$$= \begin{pmatrix} \alpha y_1 + (1-\alpha)y_2 \\ \alpha \beta_1 + (1-\alpha)\beta_2 \end{pmatrix} = \begin{pmatrix} z \\ \alpha \beta_1 + (1-\alpha)\beta_2 \end{pmatrix}$$

7 (b)

Let $f: \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be a quadratic form which is given by $f(x) = x^T Q x$, $Q \in \mathbb{R}^{n \times n}$, $Q = Q^T$.

Prove that f is convex on Ω if and only if $(x - y)^T Q (x - y) \geq 0$, for all $x, y \in \Omega$

7 (c)

▼ Let $f: \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be a function defined by $f(x) = x_1 x_2$. Is f convex on $\Omega = \{x : x_1 \geq 0, x_2 \geq 0\}$?

If a func $f: \mathbb{R}^n \subseteq \mathbb{R}$ is convex on Ω , then show that Ω is convex. 1(a)

= Suppose, Ω is not a convex set.

Then for some $\alpha \in (0, 1)$ there exists two points y_1 and

y_2 such that, $z = \alpha y_1 + (1-\alpha)y_2 \notin \Omega$

let, $\beta_1 = f(y_1)$, $\beta_2 = f(y_2)$

let, the point $(y_1, \beta_1)^T, (y_2, \beta_2)^T$

$$\therefore w = \alpha \begin{pmatrix} y_1 \\ \beta_1 \end{pmatrix} + (1-\alpha) \begin{pmatrix} y_2 \\ \beta_2 \end{pmatrix}$$

$$= \begin{pmatrix} \alpha y_1 + (1-\alpha)y_2 \\ \alpha \beta_1 + (1-\alpha)\beta_2 \end{pmatrix} = \begin{pmatrix} z \\ \alpha \beta_1 + (1-\alpha)\beta_2 \end{pmatrix}$$

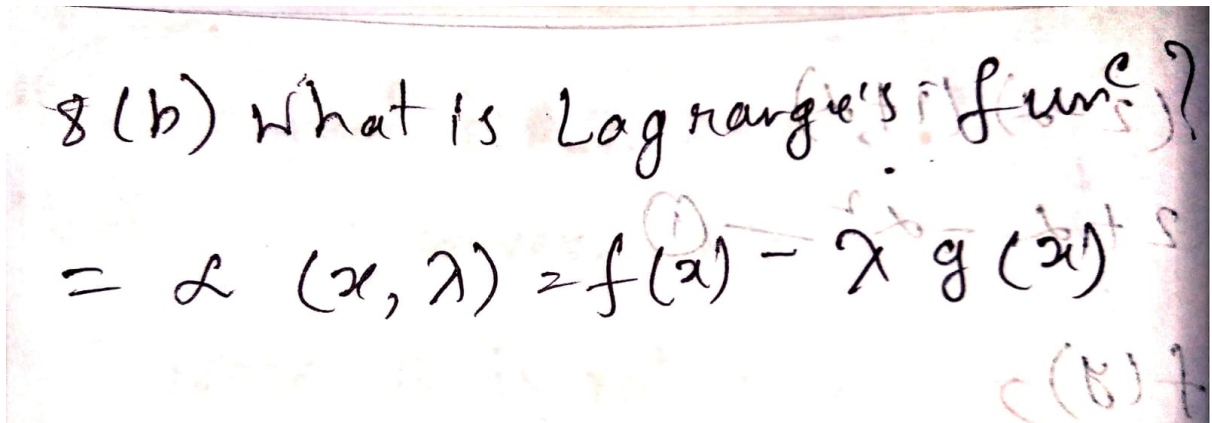
8 (a)

▼ Define unimodal function. Use Golden section search method to find the value of x that minimizes $f(x) = x^4 - 14x^3 + 60x^2 - 70x$ in the range $[0, 2]$. Locate this value of x to within a range of 0.3

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/51d22fba-2f92-4244-be7d-5314b003b918/golden_search.pdf

8 (b)

▼ Define Lagrangian function. **Prove that the Lagrange condition for a local minimizer x^* can be represented using the Lagrangian function as $Dl(x^*, \lambda^*) = 0^T$ for some λ^* , where the derivative operation D is with respect to the entire argument $(x^T, \lambda^T)^T$**



8 (c)

▼ What is a sequential quadratic programming (SQP)? **Construct SQP for the nonlinear optimization problem**

Sequential Quadratic Programming (SQP):

Sequential Quadratic Programming (SQP) is a type of iterative numerical optimization algorithm used to solve nonlinear optimization problems with equality and inequality constraints.

2020 (MATH 2111)

5 (a)

▼ Define ordinary differential equations. Solve the homogeneous differential equation $\frac{dy}{dx} = \frac{3y^2 - x^2}{2xy}$ **#review-2**

A differential equation that contains an ordinary derivative of one or more dependent variables with regard to only one independent variable is called an ordinary differential equation. For example,

$$\frac{d^2 y}{dt^2} + \frac{d^2 x}{dt^2} = x$$

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/c59aac3c-eba2-4c7f-b9d2-b5800df78d01/2020_5a.pdf

5 (b)

▼ Define Bernoulli differential equations **#2021**. Identify the differential equation $\frac{dy}{dx} + \frac{3y}{x} = 6x^2$ and solve it

A Bernoulli differential equation is a type of first-order ordinary differential equation that can be written in the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

$\frac{dy}{dx} + \frac{3y}{x} = 6x^2$ is a first order linear differential equation of the form $\frac{dy}{dx} + P(x)y = Q(x)$

where $P(x) = \frac{3}{x}$ and $Q(x) = 6x^2$

Integrating Factor:

$$\mu(x) = e^{\int P(x) dx} = e^{\int \frac{3}{x} dx} = e^{3 \ln|x|} = e^{\ln|x^3|} = x^3$$

multiply eq. (1) on both sides by $\mu(x) = x^3$, we get

$$x^3 \frac{dy}{dx} + \frac{3y}{x} (x^3) = 6x^2 (x^3)$$

$$x^3 \frac{dy}{dx} + 3x^2 y = 6x^5$$

$$\frac{d}{dx} (x^3 y) = 6x^5$$

$$\int \frac{d}{dx} (x^3 y) dx = \int 6x^5 dx$$

$$x^3 y = x^6 + C$$

$$y = \frac{x^6 + C}{x^3} \text{ is the required answer}$$

5 (c)

▼ The population x of a certain city satisfies the logistic law $\frac{dx}{dt} - \frac{1}{100}x = -\frac{1}{10^8}x^2$ where time t is measured in years. Find the population of the city at any time

1 STEP: Solve the differential equation

$$\begin{aligned}\frac{dx}{dt} &= \frac{x}{100} - \frac{x^2}{10^8} \longrightarrow \\ \frac{\frac{dx}{\left(\frac{x}{100} - \frac{x^2}{10^8}\right)}} &= dt \longrightarrow \frac{10^8 dx}{x(10^6 - x)} = dt \longrightarrow \\ 10^8 \cdot \int \frac{1}{10^6} \cdot \frac{(10^6 - x) + x}{x(10^6 - x)} dx &= \int dt \longrightarrow \\ 10^2 \cdot \int \left(\frac{1}{x} + \frac{1}{10^6 - x}\right) dx &= t \longrightarrow \\ \ln |x| - \ln |10^6 - x| &= \frac{t}{100} + \ln |C| \longrightarrow \\ \ln \left| \frac{x}{10^6 - x} \right| &= \frac{t}{100} + \ln |C| \longrightarrow \\ \frac{x}{10^6 - x} &= C \cdot e^{t/100} \longrightarrow \\ x = 10^6 \cdot C \cdot e^{t/100} - x \cdot C \cdot e^{t/100} &\longrightarrow \\ x \cdot (1 + C \cdot e^{t/100}) &= 10^6 \cdot C \cdot e^{t/100} \longrightarrow \\ x(t) &= \frac{10^6 \cdot C \cdot e^{t/100}}{1 + C \cdot e^{t/100}} \longrightarrow \\ \boxed{x(t) = \frac{10^6}{1 + \frac{1}{C} \cdot e^{-t/100}}}\end{aligned}$$

$$\boxed{x(t) = \frac{10^6}{1 + \frac{1}{C} \cdot e^{-t/100}}}$$

2 STEP: To determine the constant, we use the initial condition

$$\begin{aligned}x(1980) &= 100000 \equiv 10^5 = \frac{10^6}{1 + \frac{1}{C} \cdot e^{-1980/100}} \longrightarrow \\ 1 + \frac{1}{C} \cdot e^{-1980/100} &= 10 \longrightarrow \frac{1}{C} \cdot e^{-1980/100} = 9 \longrightarrow \\ \boxed{\frac{1}{C} = 9 \cdot e^{1980/100}}\end{aligned}$$

Conclusion,

$$x(t) = \frac{10^6}{1 + 9 \cdot e^{1980/100} \cdot e^{-t/100}} \longrightarrow$$

$$x(t) = \frac{10^6}{1 + 9 \cdot e^{(1980-t)/100}} - \text{the population at any time } t > 1980$$

3 STEP: We use the given formula to calculate the population in 2000

$$x(2000) = \frac{10^6}{1 + 9 \cdot e^{(1980-2000)/100}} = \frac{10^6}{1 + 9 \cdot e^{-0.2}} \longrightarrow$$

$$x(2000) \approx 119494.63 - \text{the population in 2000}$$

ANSWER

$$x(t) = \frac{10^6}{1 + 9 \cdot e^{(1980-t)/100}} - \text{the population at any time } t > 1980$$

$$x(2000) \approx 119494.63 - \text{the population in 2000}$$

6 (b)

▼ Define Clairaut's equation. **Find a one-parameter family of solutions of the equation $y = px + p^2$, where $p \equiv \frac{dy}{dx}$. Find singular solution, if exists, of the given equation which is not a member of the one-parameter family of solutions**

An equation of the form $y = px + f(p)$ is known as Clairaut's equation

Clairaut's equation. Definition.

[Andra 200

An equation of the form $y = px + f(p)$ is known as **Clairaut's equation**.

https://www.youtube.com/watch?time_continue=147&v=pP_ugcoTews&embeds_referring_euri=https://www.google.com/search?q=clairaut%E2%80%99s+equation&oq=Clairaut%E2%80%99s+equation&aqs=chrome.69j0j4&source_ve_path=Mjg2NjMsMjg2NjMsMjg2NjY&feature=emb_logo&ab_channel=Dr.GajendraPurohit

Clairaut's form

$$y = xp - p^2$$

$$\frac{dy}{dx} = p + x \cdot \frac{dp}{dx} - 2p \frac{dp}{dx}$$

$$p = p + \frac{dp}{dx}(x - 2p)$$

$$\frac{dp}{dx}(x - 2p) = 0$$

$$x - 2p = 0 \Rightarrow p = \frac{x}{2}$$

$$y = x \cdot \frac{x}{2} - \left(\frac{x}{2}\right)^2$$

$$= \frac{x^2}{2} - \frac{x^2}{4}$$

$$y = \frac{x^2}{4}$$

$\frac{dp}{dx} = 0$
 $p = C$
 $y = Cx - C^2$

6 (c)

Use the method of undetermined coefficients to solve the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\sin x$

▼ Answer

Sol. Given $(D^2 - 2D - 3)y = 2e^x - 10\sin x$, where $D \equiv d/dx$ (1)

Its auxiliary equation is $D^2 - 2D - 3 = 0$ so that $D = -1, 3$.

$\therefore$ C.F. $= c_1 e^{-x} + c_2 e^{3x}$, c_1, c_2 being arbitrary constants. ... (2)

Let the trial solution be $y^* = Ae^x + B\sin x + C\cos x$ (3)

Since y^* must satisfy (1), $D^2 y^* - 2D y^* - 3y^* = 2e^x - 10\sin x$ (4)

From (3), $D y^* = Ae^x + B\cos x - C\sin x$, $D^2 y^* = Ae^x - B\sin x - C\cos x$... (5)

Using (3) and (5), (4) reduces to

$$(Ae^x - B\sin x - C\cos x) - 2(Ae^x + B\cos x - C\sin x) - 3(Ae^x + B\sin x + C\cos x) = 2e^x - 10\sin x$$

or $-4Ae^x - (4B - 2C)\sin x - (4C + 2B)\cos x = 2e^x - 10\sin x$.

Equating the coefficients of like terms in above identity, we have

$$-4A = 2, \quad -(4B - 2C) = -10 \quad \text{and} \quad -(4C + 2B) = 0 \Rightarrow A = -1/2, \quad B = 2, \quad C = -1$$

$\therefore$ From (3), $y^* = (-1/2)e^x + 2\sin x - \cos x$.

and general solution is $y = \text{C.F.} + y^*$ i.e., $y = c_1 e^{-x} + c_2 e^{3x} - (1/2)e^x + 2\sin x - \cos x$.

7 (a)

▼ Define Reccati's equation. **Sate under what conditions, Reccati's equation reduces to a linear and Bernoulli's equation? Solve: $\frac{dy}{dx} + y = xy^3$ #2021**

A differential equation of the form $\frac{dy}{dx} = P(x) + Q(x)y + R(x)y^2$ where P,Q, and R are functions of x, is known as the generalized Riccati's equation.



7 (b)

▼ Solve $(D^2 + 4)y = x^2 e^{2x}$ by operator method

7 (c)

▼ Define regular singular point and examine the regular singular point of $x^2 y'' - xy' + 8(x^2 - 1)y = 0$

Definitions. A point $x = x_0$ is called an *ordinary point* of the equation

$$y'' + P(x)y' + Q(x)y = 0 \quad \dots(1)$$

if both the functions $P(x)$ and $Q(x)$ are analytic at $x = x_0$.

If the point $x = x_0$ is not an ordinary point of the differential equation (1), then it is called a *singular point* of the differential equation of (1). There are two types of singular points :

(i) **regular singular** points and (ii) **irregular singular** points.

A singular point $x = x_0$ of the differential equation (1) is called a **regular singular point** of the differential equation (1) if both $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are analytic at $x = x_0$.

A singular point, which is not regular is called an **irregular singular point**.

8 (a)

▼ Convert the differential equation $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$ with the initial conditions $y(0) = 1, y'(0) = 0$ into an integral equation. Finally, identify your obtained integral equation

8 (b)

▼ An LTI system is in zero state. Its response $y(t)$ is described by the differential equation $\frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y = \frac{df}{dt} + f(t)$. Find the response of the system if the input is given by $f(t) = 3e^{-5t}u(t)$

8 (c)

Solve the following one-dimensional heat equation by the method of separation of variables $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, 0 < x < 1, t \geq 0$ subject to the initial profile

1. $u(x, 0) = 1, 0 < x < 1$ and the Dirichlet boundary conditions

2. $u(0, t) = 0, t \geq 0$ and

3. $u(1, t) = 0, t \geq 0$

▼ Answer

2019 (MATH 2111)

5 (a)

▼ Define differential equations, its order and its degree with example #2021

5 (b)

▼ Form the differential equation of the circle represented by $y^2 - 2ay + x^2 = a^2$, a being arbitrary constant. Also, identify the differential equation

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/ee694cc4-a593-424a-a8a9-6d82ba38ba42/2019_5a.pdf

5 (c)

▼ What is first order exact differential equation? Solve $(2x \cos y + 3x^2 y)dx + (x^3 - x^2 \sin y - y)dy = 0, y(0) = 2$. #revised-1

A first-order exact differential equation is a type of ordinary differential equation (ODE) that satisfies a certain condition regarding the partial derivatives of the equation.

Consider a first-order ODE in the form:

$$\begin{aligned} M dx + N dy &= 0 & \dots (1) \\ \text{to be exact is } \partial M / \partial y &= \partial N / \partial x. & \dots (2) \end{aligned}$$

To solve the given differential equation using the method of exact differentials, we'll check if it is exact by verifying if the condition $\partial M / \partial y = \partial N / \partial x$ holds. Here, $M(x, y) = 2x \cos(y) + 3x^2 y$ and $N(x, y) = x^3 - x^2 \sin(y) - y$.

Taking the partial derivatives, we have:

$$\partial M / \partial y = -2x \sin(y) + 3x^2$$

$$\partial N / \partial x = 3x^2 - 2x \sin(y)$$

We can see that $\partial M / \partial y = \partial N / \partial x$, indicating that the differential equation is exact.

To find the solution, we need to find a potential function $\varphi(x, y)$ whose partial derivatives with respect to x and y match the given coefficients.

From $\partial\varphi/\partial x = M(x, y)$, we have:

$$\partial\varphi/\partial x = 2x\cos(y) + 3x^2y$$

Integrating with respect to x while treating y as a constant, we obtain:

$$\varphi(x, y) = x^2\cos(y) + x^3y + g(y)$$

Here, $g(y)$ is an arbitrary function of y since we integrated with respect to x .

Next, we differentiate $\varphi(x, y)$ with respect to y and equate it to $N(x, y)$:

$$\partial\varphi/\partial y = -x^2\sin(y) + 3x^3 + g'(y)$$

Comparing this with $N(x, y) = x^3 - x^2\sin(y) - y$, we can deduce that $g'(y)$ must be $-y$.

Integrating $-y$ with respect to y , we find:

$$g(y) = -y^2/2 + C$$

Here, C is an arbitrary constant of integration.

Now, we have the potential function $\varphi(x, y)$:

$$\varphi(x, y) = x^2\cos(y) + x^3y - y^2/2 + C$$

To obtain the solution of the differential equation, we set $\varphi(x, y)$ equal to a constant, which is determined by the initial condition $y(0) = 2$.

Substituting $x = 0$ and $y = 2$ into the potential function, we get:

$$\varphi(0, 2) = 0^2\cos(2) + 0^3(2) - (2^2)/2 + C$$

$$C = 2$$

Thus, the solution to the given differential equation with the initial condition $y(0) = 2$ is:

$$x^2\cos(y) + x^3y - y^2/2 = 2$$

$$(2x \cos y + 3x^2y) dx + (x^3 - x^2 \sin y - y) dy \quad (1)$$

Here, $M = 2x \cos y + 3x^2y$

$$N = x^3 - x^2 \sin y - y$$

$$\therefore \frac{\partial M}{\partial y} = 2x(-\sin y) + 3x^2$$

$$= 3x^2 - 2x \sin y$$

$$\therefore \frac{\partial N}{\partial x} = 3x^2 - 2x \sin y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Eqn (1) is in the form of exact diffⁿ eqn.

$$\begin{aligned}\therefore \int M dx &= \int 2x \cos y + 3x^2 y dx \\ &= 2 \cos y \int x dx + 3y \int x^2 dx \\ &= x^2 \cos y + x^3 y\end{aligned}$$

$$\begin{aligned}\therefore \frac{\partial}{\partial y} \left(\int M dx \right) &= x^2 (-\sin y) + x^3 \\ &= x^3 - x^2 \sin y\end{aligned}$$

$$\begin{aligned}\therefore \int \left[N - \frac{\partial}{\partial y} \left(\int M dx \right) \right] dy &= \int (x^3 - x^2 \sin y - y - x^3 + x^2 \sin y) dy \\ &= - \int y dy \\ &= - \frac{y^2}{2}\end{aligned}$$

$$\therefore \text{Sol}^n \text{ of eqn (1) is, } x^2 \cos y + x^3 y - \frac{y^2}{2} = C$$

what to do with $y(0) = 2$

6 (a)

▼ **Define homogeneous ODE #2019 #2017.** Solve the DE $(x^3 + y^2 \sqrt{x^2 + y^2})dx - xy\sqrt{x^2 + y^2}dy = 0$ **#review-**

A linear **ordinary differential equation** of order n is said to be homogeneous if it is **of the form**

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = 0, \quad (1)$$

where $y' = dy/dx$, i.e., if all the terms are proportional to a derivative of y (or y itself) and there is no term that contains a function of x alone.

or

Homogeneous equation Definition. A differential equation of first order and first degree is said to be **homogeneous** if it can be put in the form $dy/dx = f(y/x)$

$$\begin{aligned}(x^3 + y^2 \sqrt{x^2 + y^2}) dx - xy \sqrt{x^2 + y^2} dy &= 0 \\ \Rightarrow xy \sqrt{x^2 + y^2} dy &= (x^3 + y^2 \sqrt{x^2 + y^2}) dx \\ \Rightarrow \frac{dy}{dx} &= \frac{(x^3 + y^2 \sqrt{x^2 + y^2})}{xy \sqrt{x^2 + y^2}} \\ \Rightarrow \frac{d(vx)}{dx} &= \frac{(x^3 + v^2 x^2 \sqrt{x^2 + v^2 x^2})}{vx^2 \sqrt{x^2 + v^2 x^2}} \quad \text{substituting } y = vx \\ \Rightarrow v + x \frac{dv}{dx} &= \frac{(x^3 + v^2 x^3 \sqrt{1 + v^2})}{vx^3 \sqrt{1 + v^2}} \\ \Rightarrow v + x \frac{dv}{dx} &= \frac{(1 + v^2 \sqrt{1 + v^2})}{v \sqrt{1 + v^2}} \Rightarrow x \frac{dv}{dx} = \frac{(1 + v^2 \sqrt{1 + v^2})}{v \sqrt{1 + v^2}} - v \\ \Rightarrow x \frac{dv}{dx} &= \frac{1 + v^2 \sqrt{1 + v^2} - v^2 \sqrt{1 + v^2}}{v \sqrt{1 + v^2}} \\ \Rightarrow x \frac{dv}{dx} &= \frac{1}{v \sqrt{1 + v^2}} \Rightarrow v \sqrt{1 + v^2} dv = \frac{dx}{x} \\ \Rightarrow \int v \sqrt{1 + v^2} dv &= \int \frac{dx}{x} \Rightarrow \frac{1}{2} \int \sqrt{1 + v^2} (2v) dv = \int \frac{dx}{x} \\ \Rightarrow \frac{1}{3} (1 + v^2)^{\frac{3}{2}} &= \ln x + \ln c \Rightarrow (1 + v^2)^{\frac{3}{2}} = 3 \ln x + 3 \ln c \\ \Rightarrow \left(1 + \frac{y^2}{x^2}\right)^{\frac{3}{2}} &= 3 \ln |x| + 3 \ln(c) \Rightarrow \left(1 + \frac{y^2}{x^2}\right)^{\frac{3}{2}} = \ln |x^3| + \ln(c^3) \\ \Rightarrow \frac{(x^2 + y^2)^{\frac{3}{2}}}{x^3} &= \ln |x^3 c^3| \Rightarrow (x^2 + y^2)^{\frac{3}{2}} = x^3 \ln |x^3 C|\end{aligned}$$

or

We can write the given differential equation as follows,

$$x^3 dx - y\sqrt{x^2 + y^2}(xdy - ydx) = 0$$

dividing both sides by x^4 , we get the following,

$$\begin{aligned} \frac{dx}{x} - \frac{y}{x} \sqrt{1 + \left(\frac{y^2}{x^2}\right)} \frac{xdy - ydx}{x^2} &= 0 \\ \Rightarrow \frac{dx}{x} - \frac{1}{2} \sqrt{1 + \left(\frac{y^2}{x^2}\right)} d\left\{\left(\frac{y^2}{x^2}\right)\right\} &= 0 \end{aligned}$$

Integrating both sides, we get the following,

$$\Rightarrow \int \frac{dx}{x} - \frac{1}{2} \int \sqrt{1 + \left(\frac{y^2}{x^2}\right)} d\left\{\left(\frac{y^2}{x^2}\right)\right\} = c$$

$$\log |x| - \frac{1}{3} \left(1 + \frac{y^2}{x^2}\right)^{\frac{3}{2}} = c$$

$$x^3 \log |x|^3 = (x^2 + y^2)^{\frac{3}{2}} + cx^3$$

6 (b)

▼ Define Bernoulli differential equation #2021. Solve the DE $\frac{dy}{dx} + y = xy^3$ #2021

An equation of the form $(dy/dx) + Py = Qy^n$... (1A)

where P and Q are constants or functions of x alone (and not of y) and n is constant except 0 and 1, is called a **Bernoulli's differential equation**.

6 (c)

▼ $y + px = x^4 p^2$, where $p \equiv \frac{dy}{dx}$ #2019 #2017

Solve $y + px = x^4 p^2$

we have $y + px = x^4 p^2$ — (1)

Differentiating with respect to 'x' and denoting $\frac{dy}{dx}$ by 'p'

$$\frac{dy}{dx} + \frac{d}{dx}(px) = \frac{d}{dx}(x^4 p^2)$$

$$p + p \cdot 1 + x \frac{dp}{dx} = p^2 \cdot 4x^3 + x^4 \cdot 2p \frac{dp}{dx}$$

$$2p + x \frac{dp}{dx} = 4p^2 x^3 + 2p x^4 \frac{dp}{dx}$$

$$2p - 4p^2 x^3 = -x \frac{dp}{dx} + 2p x^4 \frac{dp}{dx}$$

$$2p(1 - 2px^3) = -x \frac{dp}{dx}(1 - 2px^3)$$

$$2p = -x \frac{dp}{dx}$$

$$2 \frac{dx}{x} = - \frac{dp}{p}$$

$$2p = -x \frac{dp}{dx}$$

$$2 \frac{dx}{x} = - \frac{dp}{p}$$

$$2 \frac{dx}{x} + \frac{dp}{p} = 0$$

$$2 \log x + \log p = \log c$$

$$\log x^2 + \log p = \log c$$

$$\log x^2 p = \log c$$

$$x^2 p = c$$

$$\boxed{p = \frac{c}{x^2}} \text{ — (11)}$$

$$y + px = x^4 p^2$$

$$y + \frac{c}{x^2} \cdot x = x^4 \frac{c^2}{x^4}$$

$$2px + y + px = x^4 p^2$$

differentiating with respect to 'x', denoted by 'p' — (1)

$$\begin{aligned}
 2P &= -x \frac{dP}{dx} \\
 2 \frac{dP}{P} &= - \frac{dx}{x} \\
 2 \frac{dP}{P} + \frac{dx}{x} &= 0 \\
 2 \log x + \log P &= \log C \\
 \log x^2 + \log P &= \log C \\
 \log x^2 P &= \log C \\
 x^2 P &= C \\
 \boxed{P = \frac{C}{x^2}} &\rightarrow (11)
 \end{aligned}$$

$y + Px = x^4 P^2$
 $y + \frac{C}{x^2} \cdot x = x^4 \frac{C^2}{x^4}$
 $y + \frac{C}{x} = C^2$
 $y = C^2 - \frac{C}{x}$

by putting x at x with y and P in (11) we have $y + Px = x^4 P^2$
 $y + \frac{C}{x} = C^2$
 $y = C^2 - \frac{C}{x}$

$\boxed{yx = -C + x^2 C^2}$

7 (a)

▼ Solve $y'' - 6y' + 25 = 0$, $y(0) = -3$, $y'(0) = -1$

Find the general solution
 $y'' - 6y' + 25y = 0$

Step 1: Write the characteristic Equation

$$\lambda^2 - 6\lambda + 25 = 0$$

$$\lambda^2 - 6\lambda + 25 = 0$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-6) \pm \sqrt{36 - 4(1)(25)}}{2(1)}$$

$$= \frac{6 \pm \sqrt{-64}}{2} = \frac{6 \pm 8i}{2} = 3 \pm 4i$$

$ay'' + by' + cy = 0$

Different Types of Roots	General Solution
Two Distinct Roots	$y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$
Single Root (Repeated)	$y = C_1 e^{rx} + C_2 x e^{rx}$
Complex Root ($\alpha + \beta i$)	$y = e^{\alpha x} [C_1 \cos(\beta x) + C_2 \sin(\beta x)]$

$\alpha = 3, \beta = 4i$

Step 2: General Solution

$$y = e^{3x} [C_1 \cos(4x) + C_2 \sin(4x)]$$

ans

Complex Root ($\alpha + \beta i$)	$y = e^{\alpha x} [C_1 \cos(\beta x) + C_2 \sin(\beta x)]$
-------------------------------------	--

7 (b)

▼ Solve $(D^4 + 2D^3 - 3D^2)y = 3e^{2x} + 4\sin x$ by operator method

solution of $(D^4 + 2D^3 - 3D^2)y = x^2 + 3e^{2x} + 4\sin x$

Auxiliary equation is $D^4 + 2D^3 - 3D^2 = 0$ (1)

Solving it, we get $D^2(D - 1)(D + 3) = 0$

$$D = 0, 0, 1, -3$$

Then $y = c_1 + c_2x + c_3e^x + c_4e^{-3x}$

For particular integral,

For x^2 ,

Let $y = ax^4 + bx^3 + cx^2 + dx + e$ be it's solution

then

$$y'' = 12ax^2 + 6bx + 2c$$

$$y''' = 24ax + 6b$$

$$y'''' = 24a$$

Putting values in equation $D^4 + 2D^3 - 3D^2 = x^2$, we get

$$-36ax^2 + (48a - 6b)x + (24a + 12b - 6c) = x^2$$

comparing powers,

$$\text{we get } a = -\frac{1}{36}, b = -\frac{2}{27}, c = -\frac{7}{27}$$

then P.I. for x^2 is, $-\frac{1}{36}x^4 - \frac{2}{27}x^3 - \frac{7}{27}x^2$

P.I. for $3e^{2x}$,

$$\frac{1}{D^4 + 2D^3 - 3D^2} e^{2x} = e^{2x} \frac{1}{2^4 + (2 \times 2^3) - (3 \times 2^2)} = \frac{3}{20} e^{2x}$$

P.I. for $4\sin(x)$

$$\frac{1}{D^4 + 2D^3 - 3D^2} 4\sin(x) = \frac{1}{D^2(D^2 + 2D - 3)} 4\sin(x)$$

$$= \frac{1}{(-1^2)(-1^2 + 2D - 3)} 4\sin(x) = \frac{1}{2 - D} 2\sin(x) = -2 \frac{(D+2)}{D^2 - 4} \sin(x)$$

$$= -2 \frac{\cos(x) + 2\sin(x)}{-1 - 4} = \frac{2}{5} (\cos(x) + 2\sin(x))$$

Solution of the differential equation is

$$y = c_1 + c_2x + c_3e^x + c_4e^{-3x} - \frac{1}{36}x^4 - \frac{2}{27}x^3 - \frac{7}{27}x^2 + \frac{3}{20}e^{2x} + \frac{2}{5}(\cos(x) + 2\sin(x))$$

7 (c)

- ▼ Find a series solution of $2xy'' - xy' + (x-5)y = 0$ in some interval $0 < x < R$ using the method of Frobenius

8 (a)

- ▼ Solve $p \cos(x+y) + q \sin(x-y) = z$ by Lagrange method

8 (b)

- ▼ Eliminate a, b from $z = (x^2 + a)(y^2 + b)$

→ Form the P.D.E by Eliminating the arbitrary constants from $Z = (x^2 + a)(y^2 + b)$.

Sol:- Given that $Z = (x^2 + a)(y^2 + b) \rightarrow \text{①}$

Diff. ① Partially w.r.t x .

$$\Rightarrow \frac{\partial Z}{\partial x} = (y^2 + b)(2x + 0)$$

$$\Rightarrow \boxed{\frac{p}{2x} = y^2 + b} \rightarrow \text{②}$$

Diff. ① Partially w.r.t y .

$$\Rightarrow \frac{\partial Z}{\partial y} = (x^2 + a)(2y + 0)$$

$$\Rightarrow \boxed{\frac{q}{2y} = x^2 + a} \rightarrow \text{③}$$

From ①,

$$\Rightarrow Z = \left(\frac{q}{2y}\right) \cdot \left(\frac{p}{2x}\right) \left[\frac{\text{②}}{\text{③}}\right]$$

$$\Rightarrow 4xyz = pq$$

$$\Rightarrow \boxed{pq = 4xyz}$$

Also, $\frac{\partial Z}{\partial x} = p$
 $\frac{\partial Z}{\partial y} = q$
 $(x^2)' = 2x$

8 (c)

- ▼ Apply Charpit's method to solve $p = (qy + z)^2$

Reference from book, Solve of $q = (z + px)^2$

Sol. (i). Here given equations is

$$f(x, y, z, p, q) = (z + px)^2 - q = 0 \quad \dots (1)$$

Charpit's auxiliary equations are

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

or $\frac{dp}{2p(z + px) + 2p(z + px)} = \frac{dq}{2q(z + px)} = \frac{dz}{-2px(z + px) + q} = \frac{dx}{-2x(z + px)} = \frac{dy}{0}$, by (1)

Taking the second and fourth fractions,

$$(1/q)dq = -(1/x)dx.$$

Integrating, $\log q = \log a - \log x$ so that

$$q = a/x. \quad \dots (2)$$

Substituting the above value of q in (1), we have

$$(z + px)^2 = a/x \quad \text{or} \quad px = \sqrt{a}/\sqrt{x} - z \quad \text{or} \quad p = \sqrt{a}/x\sqrt{x} - z/x. \quad \dots (3)$$

$$\therefore dz = p dx + q dy = \left(\frac{\sqrt{a}}{x\sqrt{x}} - \frac{z}{x} \right) dx + \frac{a}{x} dy, \text{ by (2) and (3)}$$

or $x dz = \sqrt{a} x^{-1/2} dx - z dx + a dy$ or $x dz + z dx = \sqrt{a} x^{-1/2} dx + a dy$

or $d(xz) = \sqrt{a} x^{-1/2} dx + a dy.$

Integrating, $xz = 2\sqrt{a}\sqrt{x} + ay + b$, a, b being arbitrary constants

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/9c5bf83c-2f0e-40c4-aaaf-a2d9606f14c3/2019_8c.pdf

2018 (MATH 2111)

5 (a)

▼ Find the differential equations of all circles which have their centres on x axis and have a given radius

644855577

Find the differential equation of those circles whose centres lie on x -axis and whose radii are variable ' r '

Soln General equation of circle

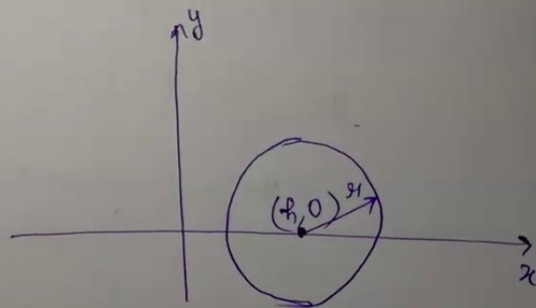
$$(x-h)^2 + (y-k)^2 = r^2$$

$(h, k) \rightarrow$ centre

$r \rightarrow$ radius,

$$(x-h)^2 + (y-0)^2 = r^2$$

$$(x-h)^2 + y^2 = r^2$$



doubtnut

$(h, k) \rightarrow$ centre

$r \rightarrow$ radius,

$$(x-h)^2 + (y-0)^2 = r^2$$

$$(x-h)^2 + y^2 = r^2$$

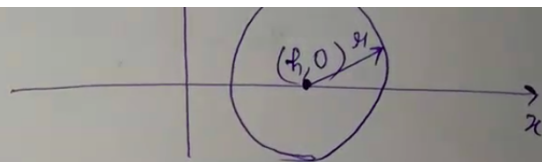
diff w.r.t x .

$$2(x-h)(1) + 2y \frac{dy}{dx} = 0$$

$$2x - 2h + 2y \frac{dy}{dx} = 0$$

$$x(x-h) = -y \frac{dy}{dx}$$

$$\frac{d}{dx} (x)^n = nx^{n-1}$$



doubtnut

$$\begin{aligned}
 x(x-h) &= -xy \frac{dy}{dx} \\
 x-h &= -y \frac{dy}{dx} \\
 \text{diff w.r.t } x & \quad \boxed{\frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}} \\
 1-0 &= -\left[y \frac{dy}{dx^2} + \frac{dy}{dx} \frac{dy}{dx} \right] \\
 1 &= -y \frac{dy}{dx^2} - \left(\frac{dy}{dx} \right)^2 \\
 y \frac{dy}{dx^2} + \left(\frac{dy}{dx} \right)^2 + 1 &= 0
 \end{aligned}$$

doubtnut

5 (b)

▼ Solve the differential equation: $\frac{dy}{dx} = x^3 y^3 - xy$ #2021

5 (c)

▼ Prove that the differential equation $(\sin x \cos y + e^{2x})dx + (\cos x \sin y + \tan y)dy = 0$ is exact and solve it #revised-1

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/2510f547-d582-45e7-b440-66d5872074ee/2018_5c.pdf

Solve according to how sir showed,

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/aced451d-817f-44c9-ac8f-f745d58debcd/2018_5c_as_sir.pdf

6

Solve the following differential equations:

▼ $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + y = \sin 2x$

for reference following solve $D^2 y + Dy + y = \sin 2x$,

Solve: $d^2y/dx^2 + dy/dx + y = \sin 2x$ or $(D^2 + D + 1)y = \sin 2x$

▶ https://www.youtube.com/watch?v=lpWFQQcni4&ab_channel=DoctorofMathematics

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$$

▼ $(D^3 - 7D - 6)y = e^{2x}x^2$

$(D^3 - 7D + 6)y = e^{2x}$ #HigherOrderLinearDifferentialEquation L760
#complementaryfunction #particularintegral #chinnaiahkalpana #mathspulse

Hello, people!

▶ https://www.youtube.com/watch?v=NurEXsbCp4Q&ab_channel=MathsPulse-ChinnaiahKalpana

Higher Order Linear Differential Equation

$$(D^3 - 7D + 6)y = e^{2x}$$

▼ $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4$

The given equation is

$$x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4 \quad \dots(1)$$

Substitute $x = e^z$ or $z = \log x$

$$\text{So that } x \frac{dy}{dx} = Dy, \quad x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

The given equation reduces to

$$\begin{aligned} D(D-1)y - 2Dy - 4y &= (e^z)^4 \\ [D(D-1) - 2D - 4]y &= e^{4z} \\ \text{i.e., } (D^2 - 3D - 4)y &= e^{4z} \quad \dots(2) \end{aligned}$$

which is an equation with constant coefficients

$$\text{A.E. is } m^2 - 3m - 4 = 0$$

$$\text{i.e., } (m-4)(m+1) = 0$$

$$\therefore m = 4, -1$$

$$\text{C.F. is } C.F. = C_1 e^{4z} + C_2 e^{-z}$$

$$\text{P.I.} = \frac{1}{D^2 - 3D - 4} e^{4z} \quad D \rightarrow 4$$

$$= \frac{1}{(4)^2 - 3(4) - 4} e^{4z} \quad Dr = 0$$

$$= \frac{1}{2D-3} z e^{4z} \quad D \rightarrow 4$$

$$= \frac{1}{(2)(4)-3} z e^{4z}$$

$$= \frac{1}{5} z e^{4z}$$

$\therefore$ The general solution of (2) is

$$y = \text{C.F.} + \text{P.I.}$$

$$y = C_1 e^{4z} + C_2 e^{-z} + \frac{1}{5} z e^{4z}$$

Substituting $e^z = x$ or $z = \log x$, we get

$$y = C_1 x^4 + C_2 x^{-1} + \frac{1}{5} \log x (x^4)$$

$$y = C_1 x^4 + \frac{C_2}{x} + \frac{x^4}{5} \log x$$

is the general solution of the Eqn. (1).

7 (a)

▼ Solve the differential equation by the method of variation of parameters: $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$

Sol. (i) Re-writing the given equation, $y_2 + (1/x) y_1 - (1/x^2) y = e^x$... (1)

Comparing (1) with $y_2 + P y_1 + Q y = R$, here $R = e^x$

Consider $y_2 + (1/x) y_1 - (1/x^2) y = 0$ or $(x^2 D^2 + xD - 1) y = 0$, $D \equiv d/dx$... (2)

Let $x = e^z$, $\log x = z$ and $D_1 \equiv d/dz$... (3)

Then $x D = D_1$ and $x^2 D^2 = D_1(D_1 - 1)$. Now, (2) becomes

$\{D_1(D_1 - 1) + D_1 - 1\} y = 0$ or $(D_1^2 - 1) y = 0$... (4)

Auxiliary equation of (4) is $D_1^2 - 1 = 0$ so that $D_1 = 1, -1$

$\therefore$ C.F. of (1) = $C_1 e^z + C_2 e^{-z} = C_1 x + C_2 x^{-1}$, as $x = e^z$... (5)

Let $u = x$ and $v = x^{-1}$. Also, here $R = e^x$... (6)

Here $W = \begin{vmatrix} u & v \\ u_1 & v_1 \end{vmatrix} = \begin{vmatrix} x & x^{-1} \\ 1 & -x^{-2} \end{vmatrix} = -2x^{-1} = -\frac{2}{x} \neq 0$... (7)

Then P.I. of (1) = $u f(x) + v g(x)$,

where $f(x) = -\int \frac{vR}{W} dx = -\int \frac{x^{-1} e^x}{(-2x^{-1})} dx = \frac{1}{2} e^x$, by (6) and (7)

and $g(x) = \int \frac{uR}{W} dx = \int \frac{x e^x dx}{(-2x^{-1})} = -\frac{1}{2} \int x^2 e^x dx$, by (6) and (7)



7.10

Method of Variation of Parameters

$= - (1/2) \times [(x^2) \times (e^x) - (2x) \times (e^x) + (2) \times (e^x)]$, using chain rule of integration by parts

$\therefore$ P.I. of (1) = $x \times (1/2) \times e^x - x^{-1} \times (1/2) \times e^x (x^2 - 2x + 2) = e^x - x^{-1} e^x$, using (6)

Hence the general solution of (1) is $y = \text{C.F.} + \text{P.I.}$

7 (b)

▼ Solve the differential equation by the method of operator factorization: $[xD^2 + (1 - x)D - 2(1 + x)]y = e^{-x}(1 - 6x)$

8 (a)

▼ Solve the partial differential equations by Charpit's method $z^2 = pqxy$

Sol. The given equation is

$$f(x, y, z, p, q) = z^2 - pqxy = 0. \quad \dots(1)$$

Charpits's auxiliary equations are

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

3.22

Non-Linear Partial Differential Equations of Order One

or
$$\frac{dp}{-pqy + 2pz} = \frac{dq}{-pqx + 2qz} = \frac{dz}{-p(-qxy) - q(-pxy)} = \frac{dx}{qxy} = \frac{dy}{pxy}, \text{ by (1)} \quad \dots(2)$$

Each fraction of (2) =
$$\frac{x dp + p dx}{x(-pqy + 2pz) + pqxy} = \frac{y dq + q dy}{y(-pqx + 2qz) + pqxy}$$

or
$$\frac{x dp + p dx}{2pxz} = \frac{y dq + q dy}{2qyz} \quad \text{or} \quad \frac{d(xp)}{xp} = \frac{d(yq)}{yq}.$$

Integrating, $\log(xp) = \log(yq) + \log a^2$ or $xp = a^2 yq. \quad \dots(3)$

Solving (1) and (2) for p and q , $p = (az)/x$ and $q = z/(ay).$

$\therefore dz = p dx + q dy = (az/x) dx + (z/ay) dy$ or $(1/z) dz = (a/x) dx + (1/ay) dy.$

Integrating, $\log z = a \log x + (1/a) \log y + \log b$ or $z = x^a y^{1/a} b.$

8 (b)

▼ Solve the partial differential equation $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$ by Lagrange's method and hence find its integral surface containing the straight line $x + y = 0, z = 1$

Ex. 6. Solve $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$. [I.A.S. 2004; Agra 2005 ; Delhi Maths (H) 2006; M.S. Univ. T.N. 2007; Indore 2003; Meerut 2009; Purvanchal 2007]

Sol. Here **Lagrange's** subsidiary equations for given equation are

$$\frac{dx}{x(y^2 + z)} = \frac{dy}{-y(x^2 + z)} = \frac{dz}{z(x^2 - y^2)} \quad \dots(1)$$

Choosing $1/x$, $1/y$, $1/z$ as multipliers, each fraction of (1)

$$= \frac{(1/x)dx + (1/y)dy + (1/z)dz}{y^2 + z - (x^2 + z) + x^2 - y^2} = \frac{(1/x)dx + (1/y)dy + (1/z)dz}{0}$$

$$\Rightarrow (1/x)dx + (1/y)dy + (1/z)dz = 0 \quad \text{so that} \quad \log x + \log y + \log z = \log c_1$$

$$\Rightarrow \log(xyz) = \log c_1 \quad \Rightarrow \quad xyz = c_1 \quad \dots(2)$$

Choosing x , y , -1 as multipliers, each fraction of (1)

$$= \frac{xdx + ydy - dz}{x^2(y^2 + z) - y^2(x^2 + z) - z(x^2 - y^2)} = \frac{xdx + ydy - dz}{0}$$

$$\Rightarrow x dx + y dy - z dz = 0 \quad \text{so that} \quad x^2 + y^2 - 2z = c_2 \quad \dots(3)$$

$\therefore$ From (2) and (3), solution is $\phi(x^2 + y^2 - 2z, xyz) = 0$, ϕ is being an arbitrary function.

2017 (MATH 2111)

5 (a)

▼ **Define order and degree of differential equation** #2021. Find the order and degree of the differential equation $2\frac{d^3y}{dx^3} + 3\left(\frac{d^2y}{dx^2}\right)^4 + \frac{dy}{dx} + y = \sin 4x$

Here, the highest derivative present is $\frac{d^3y}{dx^3}$, so the order of the differential equation is 3, and the degree of the differential equation is 1.

5 (b)

▼ **Define homogeneous ODE.** Solve the ODE $(2xy + 3y^2)dx - (2xy + x^2)dy = 0$ #review-2

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/0a4246c0-6849-45ab-93a1-f0f1c5821b07/2017_5b.pdf

5 (c)

▼ Define exact differential equation. Solve $3x(xy - 2)dx + (x^3 + 2y)dy = 0$ #review-1

Definition. If M and N are functions of x and y , the equation $M dx + N dy = 0 \dots (1)$ is called **exact** when there exists a function $f(x, y)$ of x and y , such that

$$d[f(x, y)] = M dx + N dy, \quad \dots (2)$$

$$i.e., \quad (\partial f / \partial x) dx + (\partial f / \partial y) dy = M dx + N dy. \quad \dots (3)$$

$$3x(2y-2)dx + (x^3 + 2y)dy = 0 \quad \text{--- (1)}$$

$$M = 3x(2y-2) = 3x^2y - 6x$$

$$N = x^3 + 2y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\therefore \frac{\partial M}{\partial y} = 3x^2$$

$\therefore$ Eqⁿ (1) is in the exact diffⁿ eqⁿ form

$$\therefore \frac{\partial N}{\partial x} = 3x^2$$

$$\therefore \int M dx = \int 3x^2y - 6x dx$$

$$= 3y \int x^2 dx - 6 \int x dx$$

$$= \frac{3x^3y}{3} - 3x^2$$

$$\therefore \frac{\partial}{\partial y} \left(\int M dx \right) = x^3$$

$$\therefore \left[N - \frac{\partial}{\partial y} \left(\int M dx \right) \right] dy = \int x^3 + 2y - x^3 dy$$

$$= 2 \int y dy$$

$$= \frac{2}{2} y^2$$

$\therefore$ Solⁿ of eqⁿ (1) is,

$$\therefore x^3y - 3x^2 + \frac{2}{2} y^2 = C$$

6 (a)

▼ Define Bernoulli's equation #2021. Solve $\frac{dy}{dx} + \frac{y}{x} \log y = \frac{y}{x^2} (\log y)^2$

$$\frac{dy}{dx} + \frac{y}{x} \log y = \frac{y}{x^2} (\log y)^2$$

$$\Rightarrow \frac{1}{y (\log y)^2} \frac{dy}{dx} + \frac{1}{x \log y} = \frac{1}{x^2} \quad \text{--- (1)}$$

$$\text{Let, } \frac{1}{\log y} = t$$

$$\Rightarrow \frac{dt}{dx} = \frac{-1}{y (\log y)^2} \frac{dy}{dx}$$

$$\Rightarrow -\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

Putting values in eqn (1),

$$-\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

$$\Rightarrow \frac{dt}{dx} - \frac{t}{x} = -\frac{1}{x^2}$$

$$\therefore \text{I.F.} = e^{\int -\frac{1}{x} dx} = e^{-\log x} = e^{\log \frac{1}{x}} = \frac{1}{x}$$

How

$$\text{Now, } \frac{dt}{dx} - \frac{t}{x} = -\frac{1}{x^2}$$

$$\Rightarrow \frac{1}{x} \frac{dt}{dx} - \frac{t}{x^2} = -\frac{1}{x^3}$$

$$\Rightarrow \frac{d}{dx} \left(\frac{t}{x} \right) = -\frac{1}{x^3}$$

$$\Rightarrow \frac{t}{x} = \frac{1}{2x^2} + C$$

$$\Rightarrow \frac{1}{x \log y} = \frac{1}{2x^2} + C$$

6 (b)

Solve

▼ $y + px = p^2 x^4$ #2019

▼ $y = 2px + y^2 p^3$ where $p = \frac{dy}{dx}$

$$y = 2px + y^2 p^3 \text{ where } p = \frac{dy}{dx}$$

└ ①

$$\Rightarrow y^2 = 2pxy + y^3 p^3$$

$$\Rightarrow y^2 = x(2yp) + \frac{1}{8}(2yp)^3 \rightarrow y = px + f(p)$$

└ ②

$$\text{Let, } y^2 = v$$

$$\Rightarrow 2y \frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow 2yp = \frac{dv}{dx}$$

Putting values in eqn ①,

$$v = x \frac{dv}{dx} + \frac{1}{8} \left(\frac{dv}{dx} \right)^3 \rightarrow \text{Clairaut's form}$$

$$\Rightarrow v = xc + \frac{1}{8} c \rightarrow \text{How}$$

$$\Rightarrow y^2 = cx + \frac{c}{8}$$

or

Equation Solvable For X

Q Solve $Y = 2PX + Y^2P^3$

Solⁿ $2PX = Y - Y^2P^3$
 $2X = \frac{Y}{P} - \frac{Y^2P^3}{P}$
 $2X = \frac{Y}{P} - Y^2P^2 \quad (1)$
 Diff^l both sides w.r to Y.
 $2 \frac{dx}{dy} = \frac{P \cdot 1 - Y \cdot \frac{dP}{dY}}{P^2} - \frac{Y^2 \cdot 2P \cdot \frac{dP}{dY} - 2YP^2}{P^2}$
 $\frac{2}{P} = \frac{P}{P^2} - \frac{Y}{P^2} \frac{dP}{dY} - \frac{2PY^2 \frac{dP}{dY} - 2YP^2}{P^2}$
 $\frac{2}{P} - \frac{1}{P} + 2YP^2 = -\left(\frac{1}{P^2} + 2PY\right) Y \cdot \frac{dP}{dY}$
 $\frac{1}{P} + 2YP^2 = -\left(\frac{1+2YP^3}{P^2}\right) Y \cdot \frac{dP}{dY}$

$\left(\frac{1+2YP^3}{P}\right) = -\left(\frac{1+2YP^3}{P^2}\right) Y \cdot \frac{dP}{dY}$
 $1 = -\frac{Y}{P} \frac{dP}{dY}$
 $\frac{dY}{Y} = -\frac{dP}{P}$
 $\frac{dY}{Y} + \frac{dP}{P} = 0$
 $\int \frac{1}{Y} \cdot dY + \int \frac{1}{P} \cdot dP = 0$
 $\log Y + \log P = \log C$
 $\log YP = \log C$
 $YP = C$
 $P = \frac{C}{Y}$

Put The Value of P in Eqⁿ (1), we get-

$$2X = \frac{Y}{\frac{C}{Y}} - Y^2 \left(\frac{C^2}{Y^2}\right)$$

$$2X = \frac{Y^2}{C} - C^2$$

$2XC = Y^2 - C^3$

7 (a)

Solve

▼ $(D^3 - D^2 - 6D)y = 1 + x^2$

Auxiliary Equation, $D^3 - D^2 - 6D = 0$

Implying

$$D(D + 2)(D - 3) = 0 \quad \text{or}$$

$$D = 0, 3, -2$$

Whence

$$y_{C.F.} = (c_1 + c_2 e^{3x} + c_3 e^{-2x})$$

And

$$y_{P.I.} = \frac{1}{D^3 - D^2 - 6D} (1 + x^2)$$

$$= \frac{1}{-6D \left(1 + \frac{D}{6} - \frac{D^2}{6} \right)} (1 + x^2)$$

$$= -\frac{1}{6D} \left[1 + \left(\frac{D}{6} - \frac{D^2}{6} \right) \right]^{-1} (1 + x^2)$$

$$= -\frac{1}{6D} \left[1 - \left(\frac{D}{6} - \frac{D^2}{6} \right) + \left(\frac{D}{6} - \frac{D^2}{6} \right)^2 + \dots + \right] (1 + x^2)$$

$$= -\frac{1}{6D} \left[1 - \left(\frac{D}{6} - \frac{D^2}{6} \right) + \frac{D^2}{36} \right] (1 + x^2)$$

$$= -\frac{1}{6D} \left[(1 + x^2) - \frac{D}{6} (1 + x^2) + \frac{7}{36} D^2 (1 + x^2) \right]$$

$$= -\frac{1}{6D} \left[(1 + x^2) - \frac{x}{3} + \frac{7}{36} (2) \right] = -\frac{1}{6D} \left[x^2 - \frac{x}{3} + \frac{25}{18} \right]$$

$$= -\frac{1}{6} \left[\frac{x^3}{3} - \frac{1}{3} \frac{x^2}{2} + \frac{25}{18} x \right] = -\frac{1}{108} (6x^3 - 3x^2 + 25x)$$

Therefore $y(x) = (c_1 + c_2 e^{3x} + c_3 e^{-2x})$

$$- \frac{1}{108} (6x^3 - 3x^2 + 25x)$$

▼ $(D^2 + 4)y = \cos x$

7 (b)

- ▼ Using variation of parameters to find the general solution $4y'' - 4y' - 8y = 8e^{-t}$

8 (a)

- ▼ Define regular singular point. Find the general solution of $\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 + 2)y = 0$ in power of x about $x_0 = 0$ #2021

8 (b)

- ▼ Define Laplace transformation. Find the solution of $f''(t) + 3f'(t) + 2f(t) = 4t$ where $f(0) = f'(0) = 0$ using Laplace transformation

Laplace transform is the integral transform of the given derivative function with real variable t to convert into a complex function with variable s . The Laplace transform of $f(t)$, that is denoted by $Lf(t)$ or $F(s)$ is defined by the Laplace transform formula:

LAPLACE TRANSFORM FORMULA



$$F(s) = \int_0^{+\infty} f(t) \cdot e^{-s \cdot t} \cdot dt$$

2016 (MATH 2111)

5 (a)

- ▼ Define degree and order #2021. Find the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$

Differentiation, we get,

$$\Rightarrow 2x + 2yy' + 2g + 2fy' = 0$$

Again differentiating,

$$\Rightarrow 1 + (y')^2 + yy'' + fy'' = 0 \dots\dots(i)$$

Again differentiating,

$$\Rightarrow 2(y')y'' + y'y''' + yy''' + fy''' = 0$$

$$\Rightarrow 3y'y'' + y''' \left[y - \frac{1}{y''} - \frac{(y')^2}{y''} - y \right] = 0 \text{ (from (i))}$$

$$\Rightarrow 3y'(y'')^2 - y'''[1 + (y')^2] = 0$$

5 (b)

▼ Identify and solve $(x^2 + y^2)dx - 2xydy = 0$ homogeneous equation **#review-1**

This a homogeneous equation because their degrees are same

$$\text{Given, } (x^2 + y^2)dx - 2xydy = 0$$

$$\Rightarrow (x^2 + y^2)dx = 2xydy$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 + y^2}{2xy} \dots (i)$$

$$\text{Let } y = vx$$

$$\text{Thus, } \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\text{Thus, } v + x \frac{dv}{dx} = \frac{x^2 + (vx)^2}{2x(vx)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1 + v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2 - 2v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 - v^2}{2v}$$

$$\Rightarrow \frac{dx}{x} = \frac{2v}{1 - v^2} dv$$

$$\Rightarrow \frac{dx}{x} - \frac{2v}{1 - v^2} dv = 0 \quad \dots (ii)$$

Integrating both sides, we have

$$\log x + \log(1 - v^2) = \log C$$

$$\Rightarrow \log x(1 - v^2) = \log C$$

$$\Rightarrow x(1 - v^2) = C$$

$$\Rightarrow x \left(1 - \frac{y^2}{x^2} \right) = C$$

$$\Rightarrow x \left(\frac{x^2 - y^2}{x^2} \right) = C$$

$$\Rightarrow x^2 - y^2 = Cx$$

5 (c)

▼ **Solve by the variation of parameter:** $y'' + y = \cos^2(x)$

6

Solve the following differential equations:

▼ $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = xe^{-x}$

$$(D^2 + 2D + 2)y = xe^{-x}$$

Ans. E.g. $\Rightarrow m^2 + 2m + 2 = 0$

$$m = \frac{-2 \pm \sqrt{4-8}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$= -1 \pm i$$

C.F. $= e^{-x}(A \cos x + B \sin x)$

$$y = e^{-x}(A \cos x + B \sin x + x)$$

P.I. $= \frac{1}{D^2 + 2D + 2} e^{-x} \cdot x$

$$= e^{-x} \cdot \frac{1}{(D-1)^2 + 2(D-1) + 2} \cdot x$$

$$= e^{-x} \cdot \frac{1}{D^2 - 2D + 1 + 2D - 2 + 2} \cdot x$$

$$= e^{-x} \cdot \frac{1}{D^2 + 1} \cdot x$$

$$= e^{-x} \cdot \frac{1 + D^2}{(1 + D^2)^2} \cdot x$$

$$= e^{-x} \cdot (1 - D^2 + D^4 - \dots) x$$

$$= e^{-x} (x)$$

▼ $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 4y = e^x \cos(x)$

$$(D^2 - 2D + 4)y = e^x \cos x$$

A.E. is

$$m^2 - 2m + 4 = 0$$

$$a = 1, b = -2, c = 4$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{2 \pm \sqrt{4 - 4(1)(4)}}{2}$$

$$m = 1 \pm i\sqrt{3}$$

$$y_c = (C_1 \sin \sqrt{3} x + C_2 \cos \sqrt{3} x) e^x$$

$$P.I = \frac{1}{D^2 - 2D + 4} \cdot e^x \cos x$$

$$= \frac{e^x}{(D+1)^2 - 2(D+1) + 4} \cdot 3 \cos x$$

$$= \frac{e^x}{D^2 + 2D + 1 - 2D - 2 + 4} \cdot \cos x$$

$$= \frac{e^x}{D^2 + 3} \cdot \cos' x \quad D^2 = -1^2$$

$$= \frac{e^x \cos x}{-1^2 + 3} \Rightarrow \frac{e^x \cos x}{2}$$

$$y = C.F + P.I$$

$$y = (C_1 \sin \sqrt{3} x + C_2 \cos \sqrt{3} x) e^x + \frac{e^x \cos x}{2}$$

▼ $\frac{d^2 y}{dx^2} + a^2 y = x \cos(ax)$

$(D^2 + a^2)y = x \cos ax$ Solve Differential Equation

B.Sc. Mathematics: Differential Equation: $(D^2 + a^2)y = x \cos ax$ Solve Differential Equation

▶ https://www.youtube.com/watch?v=oateSyjh4C4&ab_channel=DoctorofMathematics

**Top Mathematics Channel
on YouTube**

For B.Tech., B.E., B.Sc., B.A., Mathematics

7

- ▼ **Define regular singular points.** Find the general solution of $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$ by series method. Test convergency of the series

8 (a)

- ▼ **Solve** $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $u(x, 0) = 3\sin(2\pi x)$, $u(0, t) = 0$, $u(1, t) = 0$ where $0 < x < 1, t > 0$ by Laplace transformation

8 (b)

- ▼ **Write down Helmholtz's equation and solve it #2021**

2015 (MATH 2111)

5 (a)

- ▼ Define variable separable equation. Find an explicit solution of the initial value problem $x^2 \frac{dy}{dx} = y - xy, y(-1) = -1$ by separating variables #review-1

2.2 Separation of variables

If in an equation, it is possible to get all the functions of x and dx to one side and all the functions of y and dy to the other, the variables are said to be separable.

Working rule to solve an equation in which variables are separable.

Step 1: Let $dy/dx = f_1(x)f_2(y)$, ... (1)

be given equation. $f_1(x)$ is a function of x alone and $f_2(y)$ is a function of y alone.

Step 2: From (1), separating variables, $[1/f_2(y)] dy = f_1(x) dx$ (2)

Step 3: Integrating both sides of (2), we have

$$\int [1/f_2(y)] dy = \int f_1(x) dx + c, \quad \dots (3)$$

Subject : _____ Date :

2015 / 5(a) $x^2 \frac{dy}{dx} = y - xy$; $y = (-1)^{-1}$

$$\Rightarrow x^2 \frac{dy}{dx} = y(1-x)$$

$$\Rightarrow \frac{1}{y} dy = \frac{1-x}{x^2} dx$$

$$\Rightarrow \int \frac{1}{y} dy = \int \frac{1}{x^2} - \frac{1}{x} dx$$

$$\Rightarrow \ln|y| = -\frac{1}{x} - \ln|x| + c \quad \text{--- (1)}$$

$$\Rightarrow \ln|1| = -\frac{1}{(-1)} - \ln|(-1)| + c$$

$$\Rightarrow 0 = 1 + c$$

$$\Rightarrow c = -1$$

Putting value in eqⁿ (1),

$$\ln|y| = -\frac{1}{x} - \ln|x| - 1 \quad (\text{Ans.})$$

5 (b)

▼ Find the general solution of the differential equation $y' + 3x^2y = x^2$. Given the largest interval over which the solution is defined. Is there any transient term in the general solution?

5 (c)

▼ Solve the equation by using an appropriate substitution $x \frac{dy}{dx} + y = \frac{1}{y^2}$

$$x \frac{dy}{dx} + y = \frac{1}{y^2}$$

$$x \frac{dy}{dx} y^2 + y^3 = 1$$

Bernoulli's Equation

$$v = y^3 \rightarrow \frac{dv}{dx} = 3y^2 \frac{dy}{dx} \rightarrow \frac{dv}{3dx} = y^2 \frac{dy}{dx}$$

$$x \frac{dv}{3dx} + v = 1$$

$$\frac{dv}{dx} + \frac{3}{x} v = \frac{3}{x}$$

Integrating Factor $\rightarrow \mu(x) = e^{\int \frac{3}{x} dx} = e^{3 \ln x} = e^{\ln x^3} = x^3$

$$\frac{dv}{dx} x^3 + \frac{3}{x} x^3 v = \frac{3}{x} x^3$$

$$\frac{dv}{dx} x^3 + 3x^2 v = 3x^2$$

$$\int \left[\frac{dv}{dx} x^3 + 3x^2 v \right] dx = 3 \int x^2 dx$$

Product Rule of Derivative

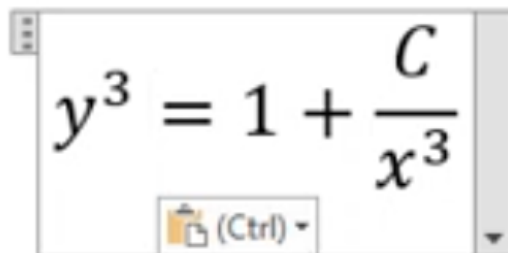
$$y' = u' v + v' u \rightarrow y = uv$$

$$\int \left[\frac{dv}{dx} x^3 + (x^3)' v \right] dx = 3 \int x^2 dx$$

$$x^3 v = 3 \frac{x^3}{3} + C$$

$$x^3 v = x^3 + C$$

$$v = 1 + \frac{C}{x^3}$$



$$y^3 = 1 + \frac{C}{x^3}$$

$$y = \sqrt[3]{1 + \frac{C}{x^3}}$$

6

Solve the following differential equations:

▼ $y'' + 4y' + 4y = 2x + 6$

▼ $y'' - y = x^2 e^x + 5$

▼ $y'' - y' - 12y = e^{4x}$